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Tamil Nadu Innovation Modernization through
Problem Solving and Advanced Creative Tech



TN-IMPACT 2026

IDENTIFYING INDUSTRIAL CHALLENGES

PROBLEM STATEMENT NUMBER : TNI26204

PROBLEM STATEMENT TITLE : Design of a Miniaturized Fluid Viscous Damper





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Team details

Team Name : FORGEX

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College Name: COIMBATORE INSTITUTE OF TECHNOLOGY

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Problem statement

Design a compact fluid viscous damper (**<100 mm length, 30 mm diameter**) capable of providing effective vibration and shock attenuation in space-constrained systems. The damper must incorporate a controlled piston-orifice mechanism, reliable sealing, and clevis-type mounting with spherical bearings for precise and durable performance.



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Problem description

- **Problem:** Vibration challenges in UAV Optical Payload Systems: UAV Payloads experience high frequency vibration (20-300 Hz).

Leads to:

1. Image blur
2. Sensor Drift
3. Mission Inaccuracy

- **Need:** Compact, high-reliability fluid viscous damper for controlled vibration attenuation within tight size constraints, boosting payload stability and surveillance performance.





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Objectives

- To design a compact and lightweight fluid viscous damper to improve the overall **imaging stability** and **reliability of optical payloads** used in **military surveillance**.
- To introduce a **segmented spiral orifice** flow path inside the damper for better control over viscous damping behaviour.
- To introduce a legitimate **standard sealing system** for arresting leaks.
- To study and verify the performance of the damper using basic analytical calculation and simulation tools.





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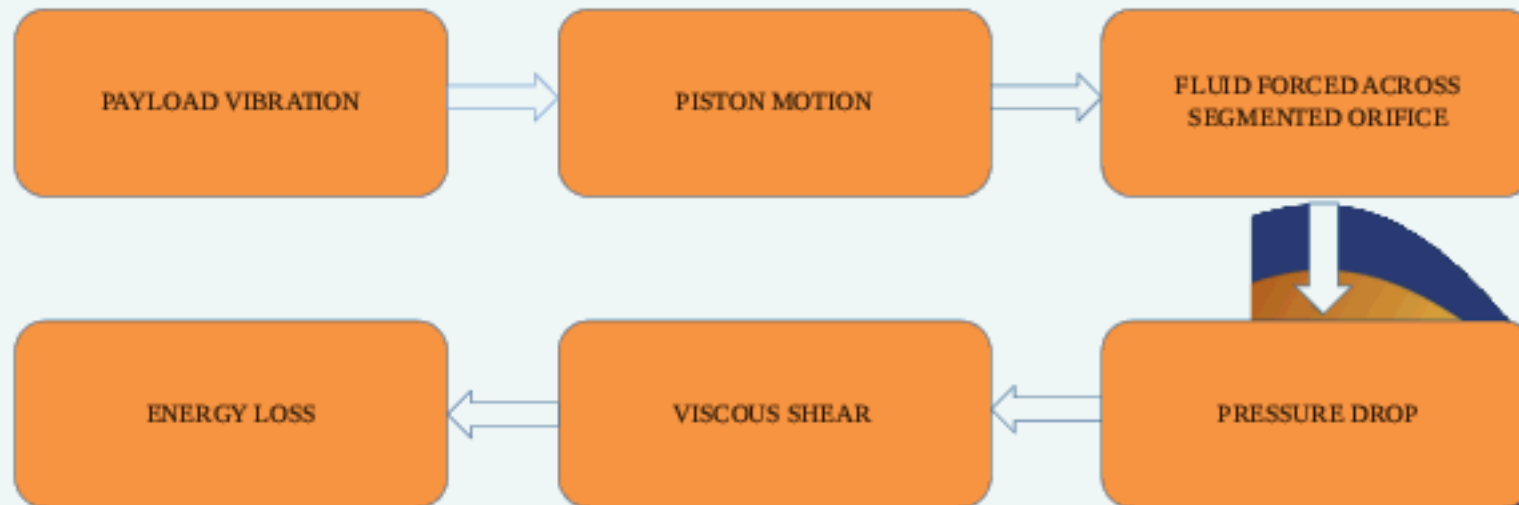
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System Working Principle

ENERGY DISSIPATION MECHANISM

EQUATION : $F = C_v$



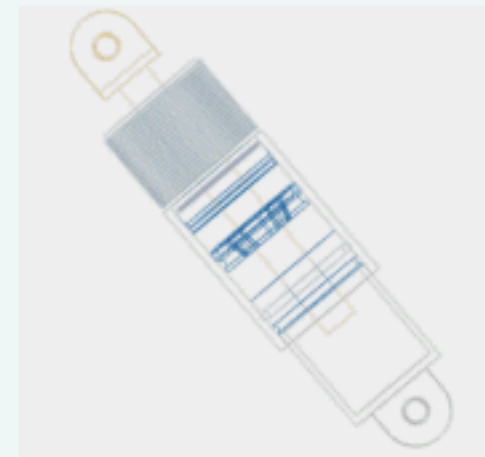
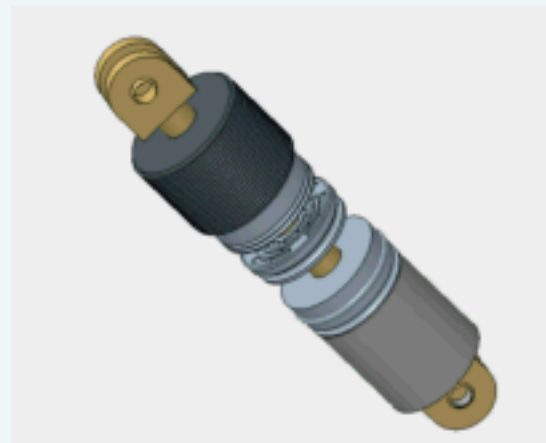
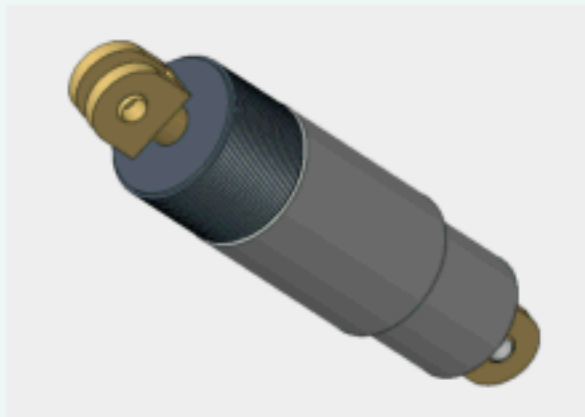


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Design Overview



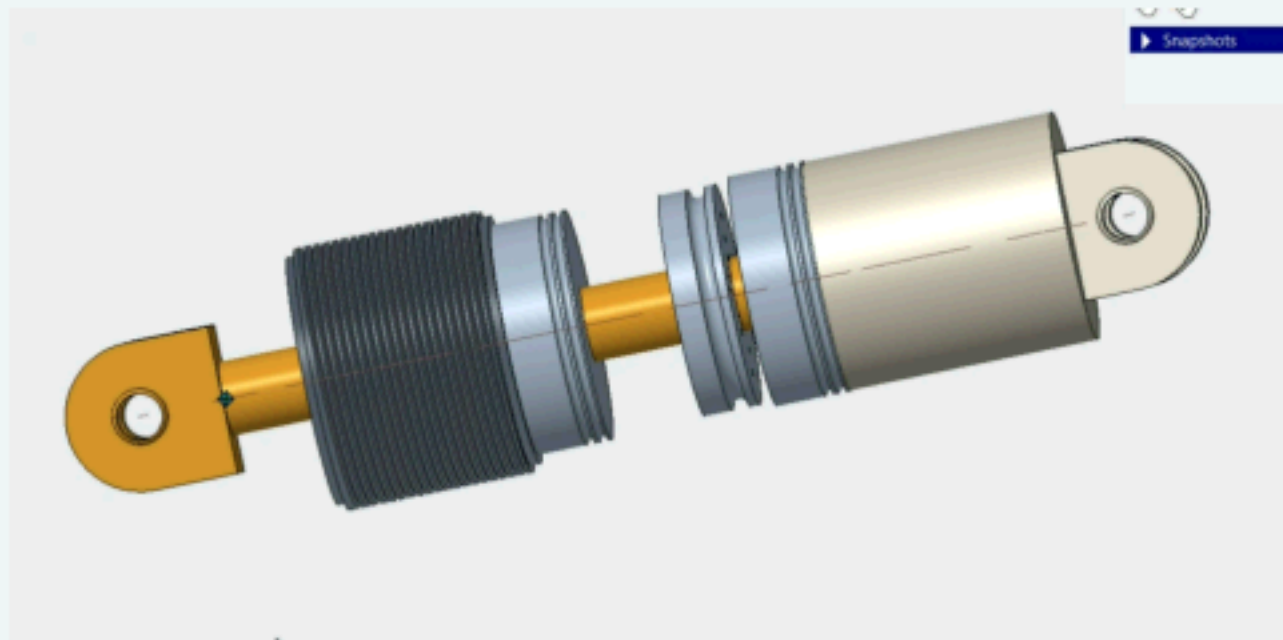


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Simulation





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Design Parameters

PARAMETER	SPECIFICATION
OVERALL LENGTH	97 mm
OVERALL DIAMETER	28.5 mm
FLUID CHAMBER LENGTH	35 mm
STROKE LENGTH	24 mm
PISTON DIAMETER	24 mm
OIL VISCOSITY	0.97 kg/ms
OIL DENSITY	970 Kg/m ³



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Material Selection

MATERIAL	COMPONENTS	REASON
ALUMINIUM	CYLINDER,, PISTON, SEAL HEAD, END CAP	LOW WEIGHT, GOOD STIFFNESS-TO-WEIGHT RATIO
STAINLESS STEEL	PISTON ROD	HIGH STRENGTH, WEAR RESISTANCE
NITRILE RUBBER	RUBBER BELLOW	EFFECTIVE SEALING, COMPATIBILITY
SILICONE OIL	DAMPING FLUID	THERMAL STABILITY, GOOD DAMPING PERFORMANCE





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Orifice System

Innovation Over Conventional Piston Designs

Parameter	Conventional Orifice	Proposed Design
Flow Path	Straight	Spiral extended path
Pressure Drop	Localized	Distributed
Damping Control	Limited	Progressive
Flow Stability	Moderate	Improved
Vibration Attenuation	Basic	Enhanced



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Orifice System

- **Innovation — Segmented Spiral Design**
- Segmented spiral flow path increases **effective viscous** travel length within a compact piston.
- Generates **distributed pressure drop**, enabling controlled damping force development.
- Offset micro-orifices provide progressive throttling, **improving stability** under varying vibration velocities.
- Enhances **high-frequency vibration attenuation** without increasing damper size.
- Enables **tunable damping characteristics** through spiral pitch and orifice geometry optimisation.



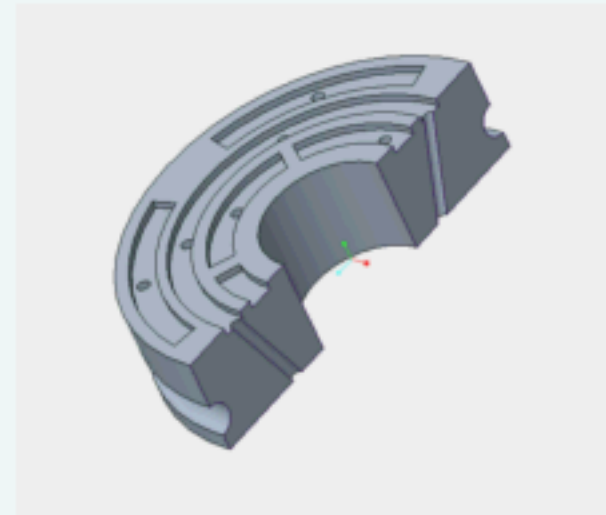
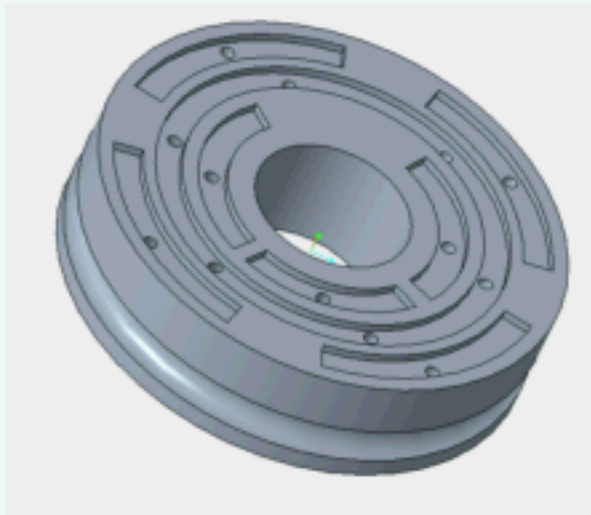


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Orifice System - Drawing





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Seal System

Sealing Requirement

Sealing has been considered to retain the damping fluid during piston reciprocation while avoiding excessive friction.

Primary Piston Seal

An elastomeric O-ring has been designed in the piston outer diameter groove to control internal fluid bypass.





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Seal System

Low-Friction Seal Provision

A PTFE grooved seal with an energized O-ring has been included along the piston guiding region to improve wear behaviour and reduce sliding resistance.

Rod Exit Seal Arrangement

A dual lip seal with has been planned at the cylinder end cap where the piston rod exits, to limit external leakage and prevent dust accumulation.





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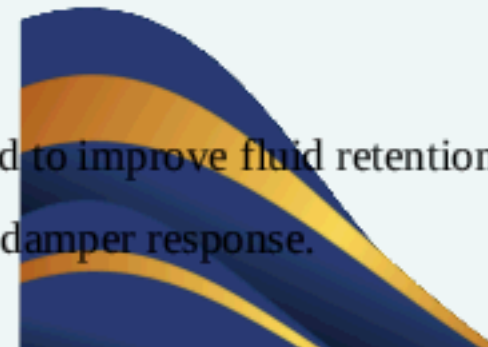
Seal System

Manufacturing Considerations in Design

- Necessary groove dimensions for seal placement have already been incorporated in the piston model.
- Cylinder bore finishing requirements have been specified to support seal life and stable damper performance.

Overall Design Understanding

Using a combination of these sealing provisions is expected to improve fluid retention while keeping friction within acceptable limits for smooth damper response.





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ORIFICE AREA RATIO CALCULATION:

Piston Area,

$$= \pi / 4 \times D^2$$

$$A_p = \pi / 4 (24)^2 = 452.39 \text{ mm}^2$$

Piston area = 452.39 mm²

AREA OF ONE MICRO-HOLE

$$= \pi / 4 \times d^2$$

$$= \pi / 4 (0.7)^2 = 0.3848 \text{ mm}^2$$

Area of one micro-hole = 0.3848 mm²





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TOTAL ORIFICE AREA

$$A_{ot} = n \times A_o$$

$$A_{ot} = 11 \times 0.3848 = 4.23 \text{ mm}^2$$

Total orifice area = 4.23 mm²

ORIFICE AREA PERCENTAGE

$$\text{Orifice area ratio} = A_p / A_{ot} \times 100$$

$$= 4.23 / 452.39 \times 100$$

$$= 0.93\%$$

Orifice area ratio = 0.93%

This is a reasonable value for a compact viscous damper.



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FLOW RATE

Convert piston area into m^2 :

$$A_p = 452.39 \times 10^{-6} = 4.5239 \times 10^{-4} \text{ m}^2$$

Flow rate:

$$Q = A_p \times V_p$$

$$Q = 4.5239 \times 10^{-4} \times 0.1$$

$$Q = 4.52 \times 10^{-5} \text{ m}^3/\text{s}$$

Flow rate = $4.52 \times 10^{-5} \text{ m}^3/\text{s}$





PRESSURE DROP:

For flow through small orifices, the pressure drop is estimated by the standard orifice relation:

$$Q = C_d A \sqrt{\frac{2\Delta P}{\rho}}$$

Rearranging for pressure drop:

$$\Delta P = \frac{\rho}{2} \left(\frac{Q}{C_d A} \right)^2$$

- Silicone oil density: $\rho = 970 \text{ kg/m}^3$
- Discharge coefficient for small sharp-edged orifices is : $C_d = 0.62$

$$\Delta P = \frac{970}{2} \left(\frac{4.52 \times 10^{-5}}{0.62 \times 4.23 \times 10^{-6}} \right)^2$$

$\Delta P = 1.13 \times 10^5 \text{ Pa}$ (Close to Analytical Value)





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DAMPING FORCE:

$$F_d = \Delta P \times A_p$$

$$F_d = 0.144 \times 106 \times 4.5239 \times 10^{-4}$$

$$F_d = 65.2 \text{ N}$$

Estimated damping force = 65.2 N

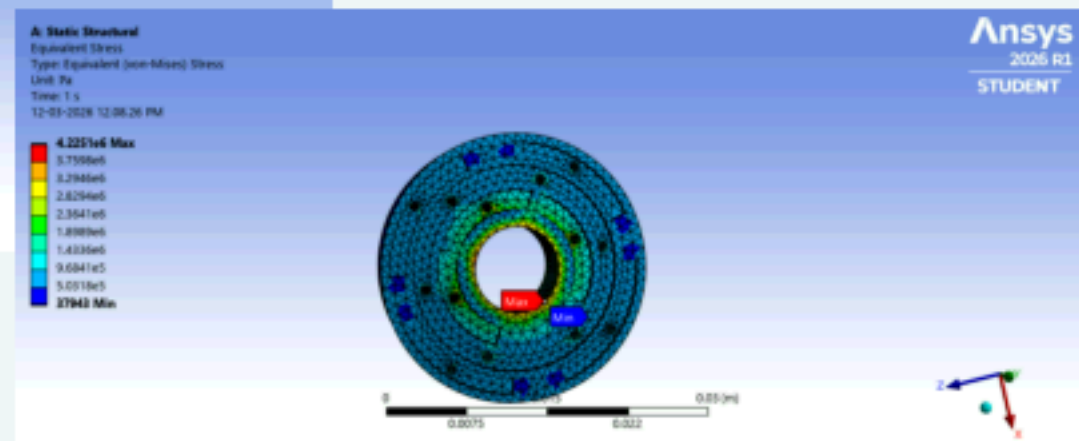
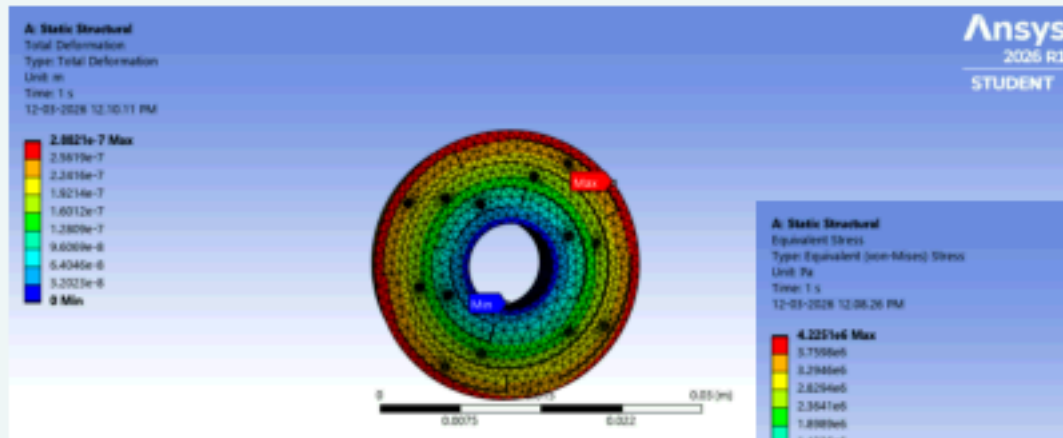




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Static Structural Analysis

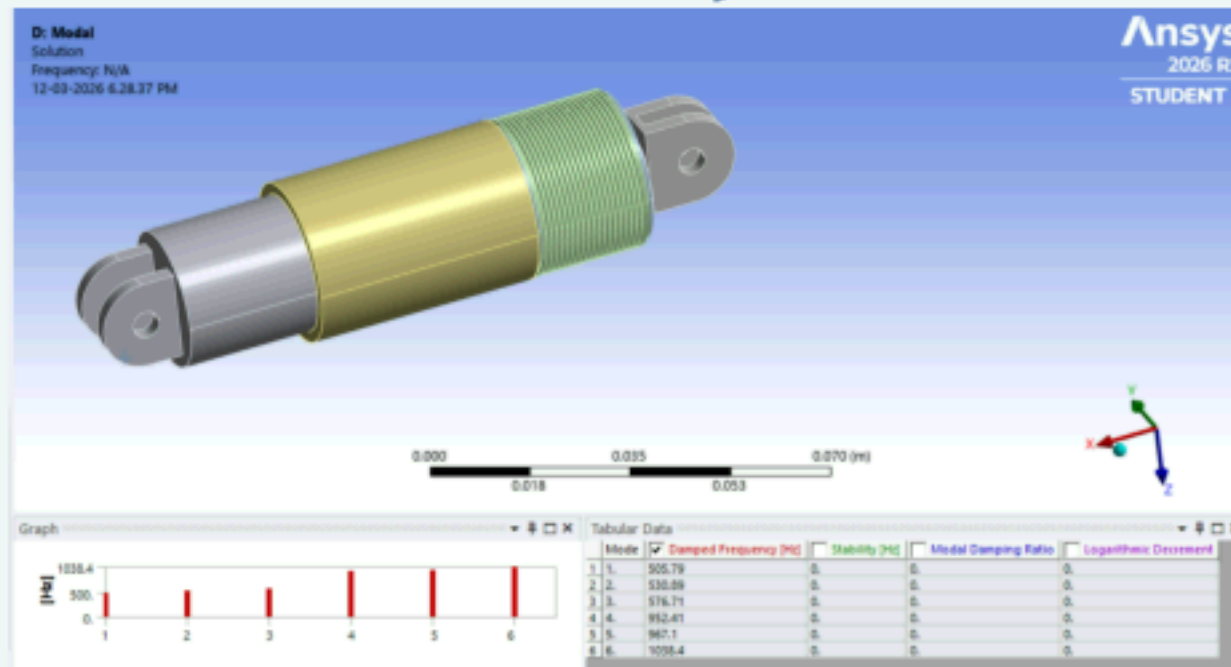




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Modal Analysis

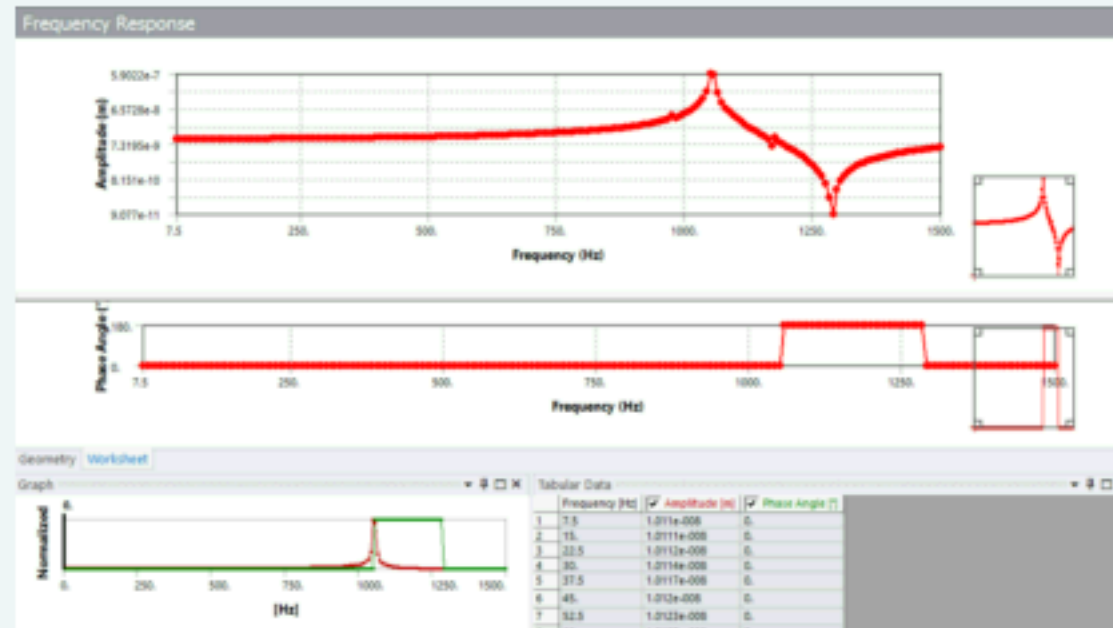




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Harmonic Analysis



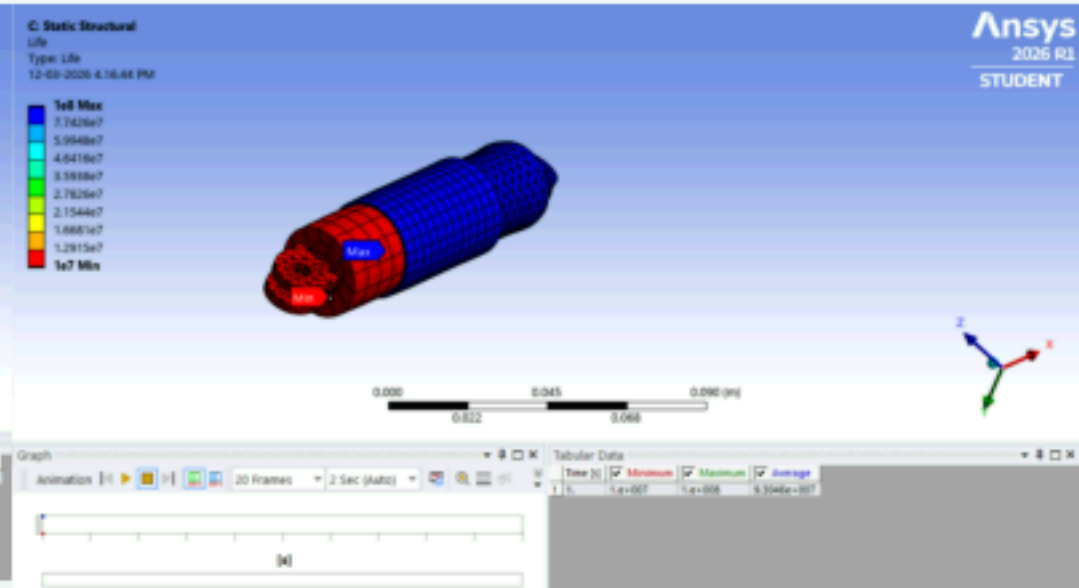
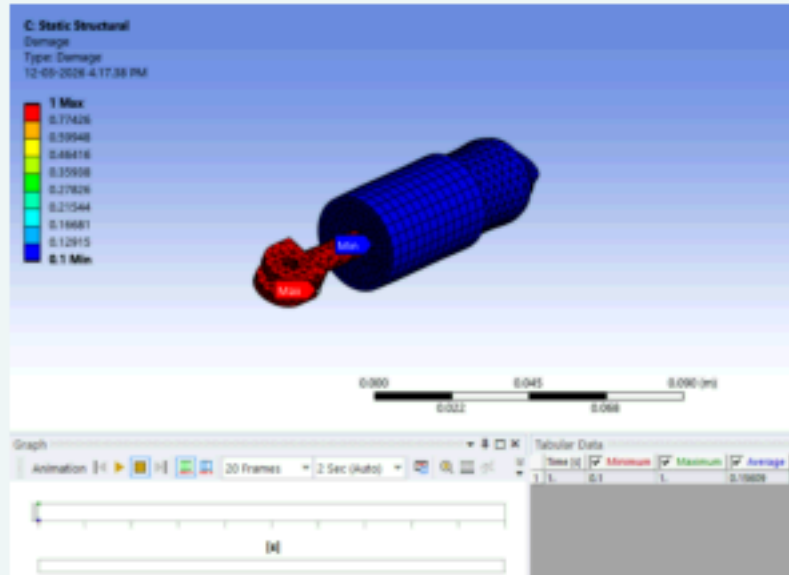


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Fatigue Life Analysis

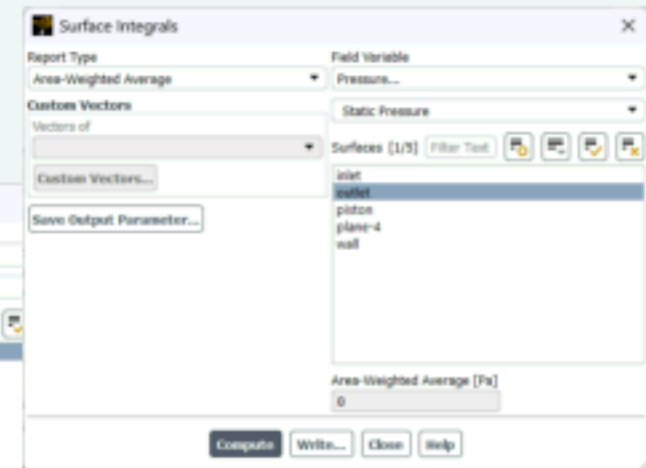
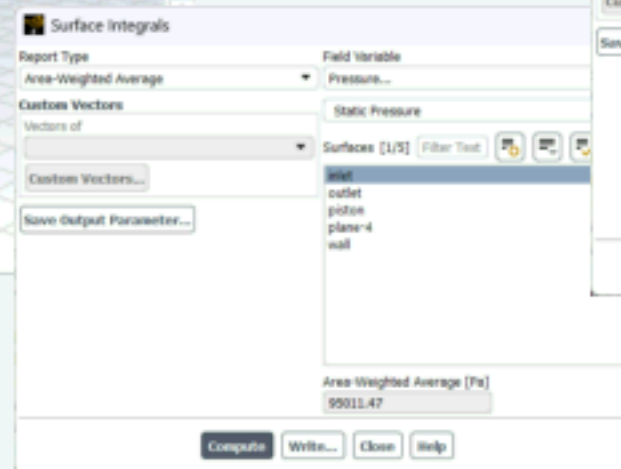
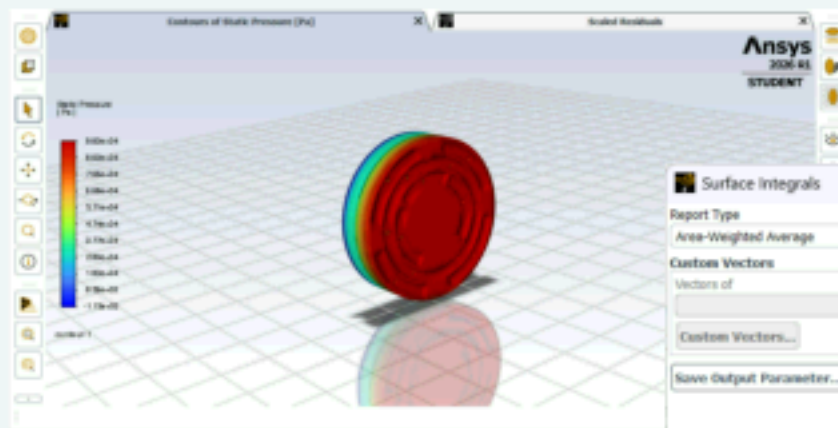




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CFD - Pressure Difference



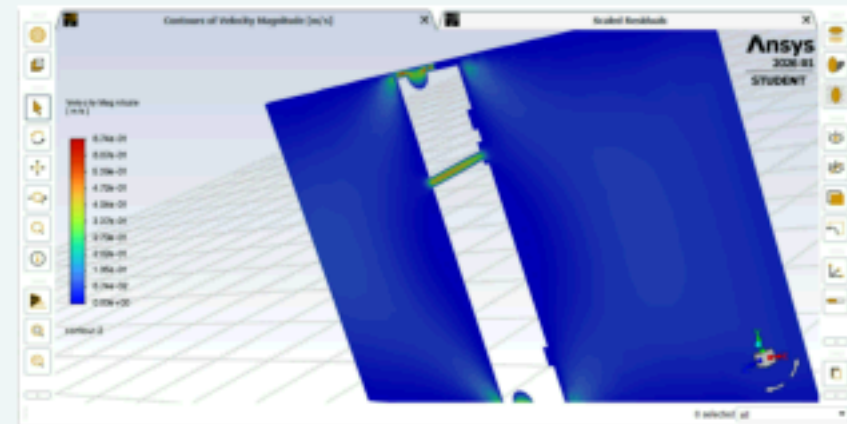
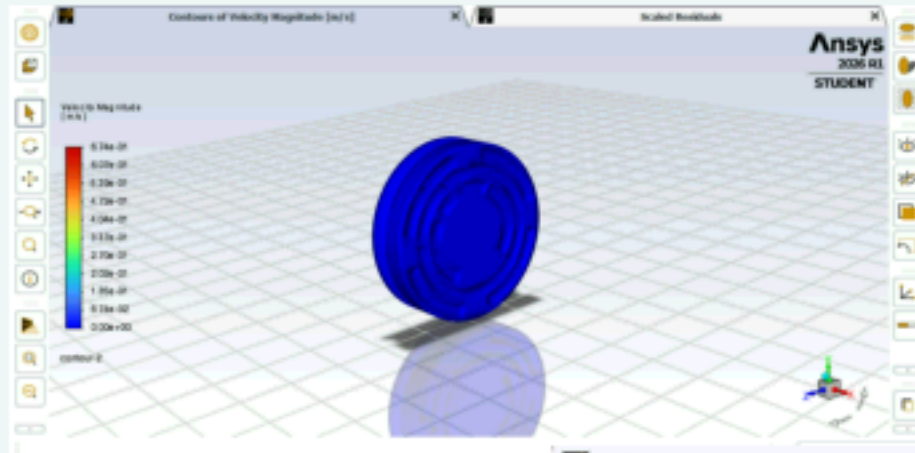


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CFD - Velocity Distribution



Volume Integrals

Report Type

- ☐ Mass-Average
- ☐ Mass Integral
- ☐ Mass
- ☐ Sum
- ☐ Minimum
- ☒ Maximum
- ☐ Volume
- ☐ Volume-Average
- ☐ Volume Integral

Field Variable

Velocity...

Velocity Magnitude

Max [m/s]

0.8891013

Save Output Parameter...

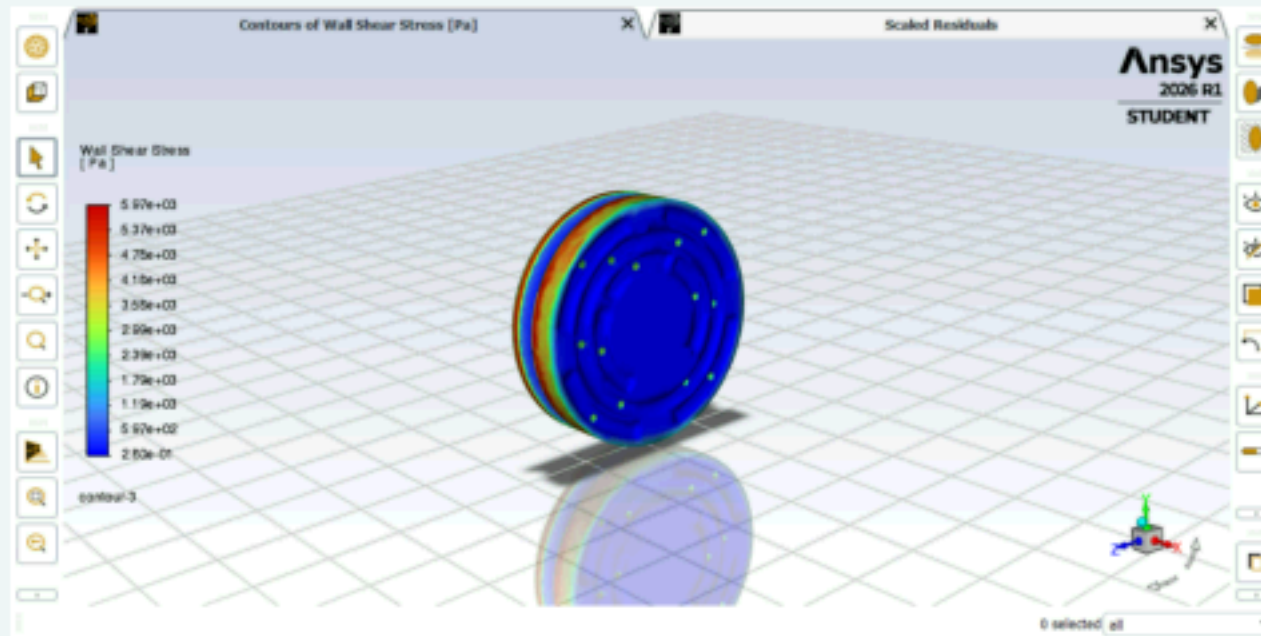


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CFD - Shear At Wall



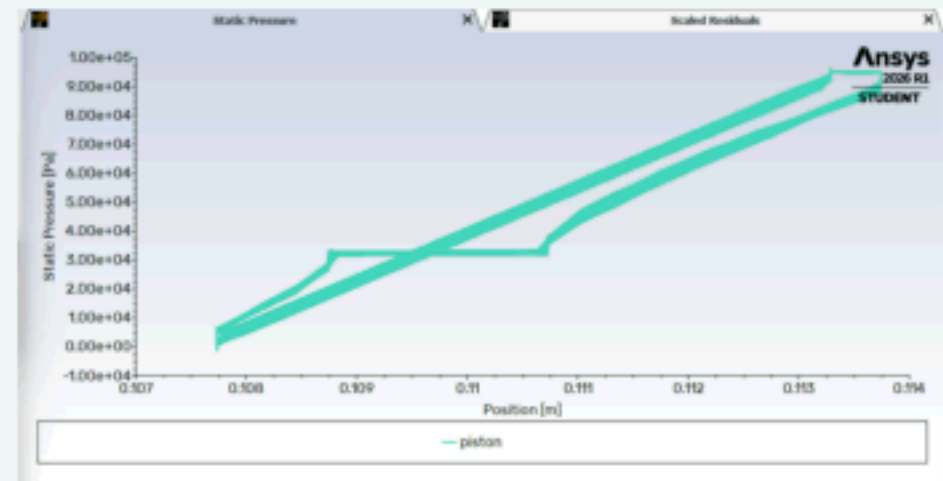
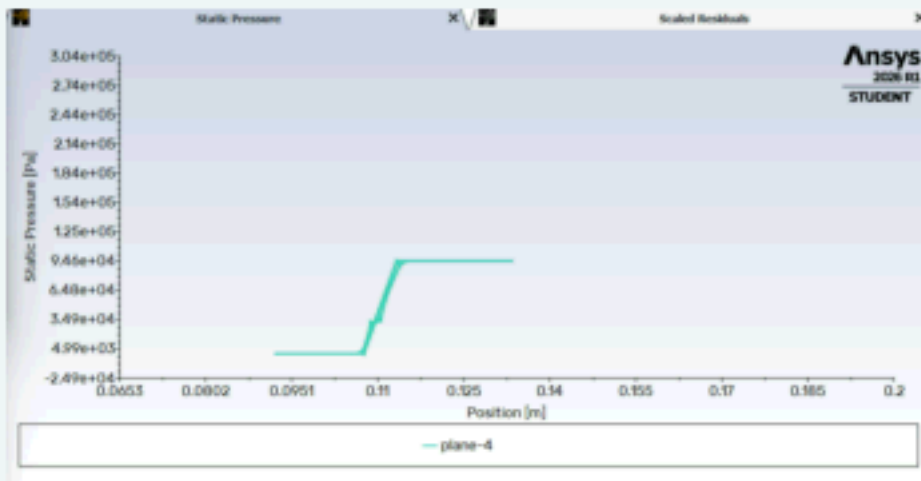


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CFD - Plots

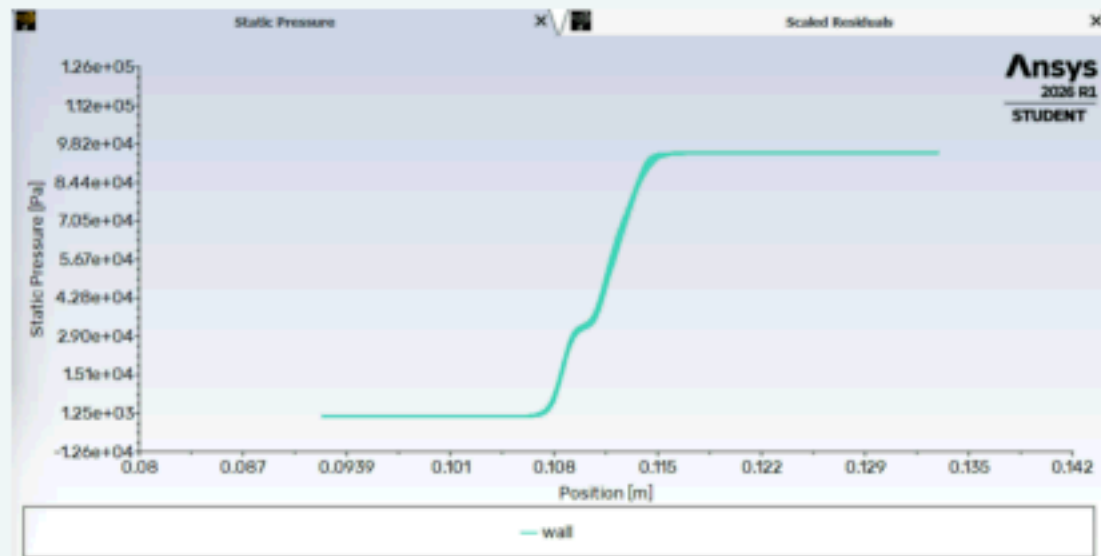




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CFD - Plots





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Total Mass

Component	Material	Mass (g)
Cylinder body	Aluminium	22
Piston	Aluminium	9
Piston rod	SS304	24
End caps (2)	Aluminium	10
Clevis pin	SS304	3
Silicone oil (19.6 ml)	Silicone oil	19
Seals (O-rings)	Nitrile rubber	4
Fasteners	Steel	3
Total		94



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Conclusion

- Structurally safe under expected UAV vibration loading
- Natural frequencies lie outside operational excitation range
- Spiral orifice produces controlled pressure drop and viscous damping
- Adequate fatigue life for repeated mission cycles
- Compact and lightweight design improves payload stability

Overall, the compact and lightweight configuration can help improve optical payload stability in UAV systems.

